

CS173: Discrete Structures

Problem Set 3

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Throughout this problem set, we will use the following shorthand notation for axioms:

- **Extensionality** refers to The Axiom of Extensionality (Axiom 1)
- **Pairing** refers to The Axiom of Pairing (Axiom 2)

1. Prove each of the following statements.

(a) Prove $\forall x \forall y \forall z ((x \subseteq y \wedge y \subseteq z) \Rightarrow x \subseteq z)$.

Proof. Let x , y , and z be arbitrary sets. Assume $x \subseteq y$ and $y \subseteq z$.

To prove $x \subseteq z$, we must show $\forall w (w \in x \Rightarrow w \in z)$ by the definition of subset.

Let w be an arbitrary set and assume $w \in x$.

Since $x \subseteq y$, we have $\forall u (u \in x \Rightarrow u \in y)$ by the definition of subset.

Thus $w \in y$.

Since $y \subseteq z$, we have $\forall u (u \in y \Rightarrow u \in z)$ by the definition of subset.

Thus $w \in z$.

Therefore $\forall w (w \in x \Rightarrow w \in z)$, which means $x \subseteq z$ by the definition of subset. $\square$

(b) Prove $\forall x \forall y (x \cap y \subseteq x)$.

Proof. Let x and y be arbitrary sets.

To prove $x \cap y \subseteq x$, we must show $\forall w (w \in (x \cap y) \Rightarrow w \in x)$ by the definition of subset.

Let w be arbitrary and assume $w \in (x \cap y)$.

Since $x \cap y$ is an intersection of x with y , we have

$\forall u (u \in x \cap y \Leftrightarrow ((u \in x) \wedge (u \in y)))$ by the definition of intersection of two sets.

Thus $w \in x \wedge w \in y$.

Therefore $w \in x$.

Therefore $\forall w (w \in x \cap y \Rightarrow w \in x)$, which means $x \cap y \subseteq x$ by the definition of subset. $\square$

(c) Prove $\forall x \forall y (x \subseteq x \cup y)$.

Proof. Let x and y be arbitrary sets.

To prove $x \subseteq x \cup y$, we must show $\forall w (w \in x \Rightarrow w \in x \cup y)$ by the definition of subset.

Let w be arbitrary and assume $w \in x$.

Consider the definition $x \cup y := \{u \mid u \in x \vee u \in y\}$.

Since $w \in x$, we have $w \in x \vee w \in y$.

Thus $w \in x \cup y$.

Therefore $\forall w (w \in x \Rightarrow w \in x \cup y)$, which means $x \subseteq x \cup y$ by the definition of subset. $\square$

(d) Prove $\forall x \forall y (x \cap y \subseteq x \cup y)$.

Proof. Let x and y be arbitrary sets.

By problem 1b, we have $x \cap y \subseteq x$.

By problem 1c, we have $x \subseteq x \cup y$.

By problem 1a (transitivity of $\subseteq$), since $x \cap y \subseteq x$ and $x \subseteq x \cup y$, we have $x \cap y \subseteq x \cup y$. $\square$

2. Prove each of the following statements.

(a) Prove $\forall x \forall y \exists p (p = (x, y))$.

Proof. Let x and y be arbitrary sets.

By Pairing applied to x and x , there exists a set $\{x, x\}$.

By Extensionality, $\{x, x\} = \{x\}$.

By Pairing applied to x and y , there exists a set $\{x, y\}$.

By Pairing applied to $\{x\}$ and $\{x, y\}$, there exists a set $\{\{x\}, \{x, y\}\}$.

Thus, by definition, there exists an ordered pair (x, y) .

Therefore $\exists p (p = (x, y))$. $\square$

(b) Prove $\forall a \forall b \forall x \forall y ((a, b) = (x, y) \Leftrightarrow (a = x \wedge b = y))$.

Proof. Let a , b , x , and y be arbitrary sets.

We prove both directions of the biconditional.

($\Rightarrow$) Assume $(a, b) = (x, y)$.

By the definition of ordered pair, $\{\{a\}, \{a, b\}\} = \{\{x\}, \{x, y\}\}$.

We consider two cases.

Case 1: Suppose $a = b$.

Since $a = b$, we have $\{a, b\} = \{a\}$ by Extensionality.

Thus $\{\{a\}, \{a, b\}\} = \{\{a\}, \{a\}\} = \{\{a\}\}$ by Extensionality.

Therefore $\{\{x\}, \{x, y\}\} = \{\{a\}\}$.

By Extensionality, both $\{x\}$ and $\{x, y\}$ must equal $\{a\}$.

From $\{x\} = \{a\}$, we have $x = a$.

From $\{x, y\} = \{a\}$, we have $y = a$.

Thus $a = x$ and $b = a = y$, so $a = x$ and $b = y$.

Case 2: Suppose $a \neq b$.

Since $a \neq b$, we have $b \in \{a, b\}$ but $b \notin \{a\}$.

Thus $\{a\} \neq \{a, b\}$ by The Extensionality.

Since $\{\{a\}, \{a, b\}\} = \{\{x\}, \{x, y\}\}$, we know $\{a\} \in \{\{x\}, \{x, y\}\}$.

By Extensionality, either $\{a\} = \{x\}$ or $\{a\} = \{x, y\}$.

Suppose $\{a\} = \{x, y\}$.

Then both x and y must equal a by Extensionality.

Thus $x = a = y$, which means $x = y$.

But then $\{x, y\} = \{x\}$ by The Extensionality.

Since $\{a, b\} \in \{\{a\}, \{a, b\}\} = \{\{x\}, \{x, y\}\}$ and

$\{x, y\} = \{x\}$, we have $\{a, b\} = \{x\}$.

Combined with $\{a\} = \{x\}$, this gives $\{a, b\} = \{a\}$.

This contradicts $\{a\} \neq \{a, b\}$.

Therefore $\{a\} \neq \{x, y\}$, so we must have $\{a\} = \{x\}$.

From $\{a\} = \{x\}$, we have $a = x$ by Extensionality.

Since $\{a, b\} \in \{\{a\}, \{a, b\}\}$ and $\{a, b\} \neq \{a\} = \{x\}$,

we have $\{a, b\} = \{x, y\}$.

From $\{a, b\} = \{x, y\}$ and $a = x$, we have $b \in \{x, y\}$.

If $b = x$, then $b = x = a$, contradicting $a \neq b$.

Thus $b = y$.

In both cases, $a = x$ and $b = y$.

($\Leftarrow$) Assume $a = x$ and $b = y$.

Since $a = x$, we have $\{a\} = \{x\}$ by Extensionality.

Since $a = x$ and $b = y$, we have $\{a, b\} = \{x, y\}$ by Extensionality.

Therefore $\{\{a\}, \{a, b\}\} = \{\{x\}, \{x, y\}\}$ by Extensionality.

By the definition of ordered pair, $(a, b) = (x, y)$.

Therefore $(a, b) = (x, y) \Leftrightarrow (a = x \wedge b = y)$. □

3. Prove each of the following statements.

(a) Prove that $\forall x(x \cup \{x\} \neq \emptyset)$.

Proof. Let x be an arbitrary set.

By Pairing applied to x and x , there exists a set $\{x, x\}$.

By Extensionality, $\{x, x\} = \{x\}$.

Since $\{x\} \subseteq x \cup \{x\}$ by Problem 1c, and $x \in \{x\}$, we have $x \in x \cup \{x\}$ by the definition of subset.

Suppose for contradiction that $x \cup \{x\} = \emptyset$. By Extensionality, $x \cup \{x\}$ and $\emptyset$ have the same elements. But $x \in x \cup \{x\}$ and $x \notin \emptyset$, a contradiction.

Therefore, $x \cup \{x\} \neq \emptyset$. □

(b) Prove that $\forall x \forall y (x \neq y \Rightarrow x \cup \{x\} \neq y \cup \{y\})$.

Proof. Let x and y be arbitrary sets such that $x \neq y$.

To prove this we will use reductio ad absurdum.

Assume $x \cup \{x\} = y \cup \{y\}$.

As shown in problem 3a, we have $x \in x \cup \{x\}$, and so $x \in y \cup \{y\}$.

So, by Extensionality, $x \in y$ or $x \in \{y\}$.

Since $x \in \{y\}$ implies, by Extensionality, $x = y$, which contradicts $x \neq y$, we have $x \in y$.

Again, as shown in problem 3a, we have $y \in y \cup \{y\}$ and so $y \in x \cup \{x\}$.

Assume, for the sake of this proof, $y \notin x$.

Since $y \in x \cup \{x\}$ and $y \notin x$, we must have $y \in \{x\}$, so $y = x$.

We have both $x \neq y$ and $y = x$, which is a contradiction.

Therefore, by reductio ad absurdum, we have $x \neq y \Rightarrow x \cup \{x\} \neq y \cup \{y\}$. □

However, by applying the same symmetric argument that showed $x \in y$, we can also derive $y \in x$. This means the step where we assumed $y \notin x$ is not justified. With only Axioms 0, 1, and 2, we cannot rule out the case where $x \in y$ and $y \in x$ simultaneously.